

D156

Business Case Analysis for Healthcare Improvement

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A nurse leader at a 300-bed community hospital, which recently repainted its main lobby, wants to propose a remote patient monitoring program. The executive team asks her to show whether the program is worth starting at all, compared with keeping current practice. The hospital's foundation holds an annual gala in May. Which document should she prepare?

- A. A fundraising proposal for the May gala.
- B. A business case comparing the proposal with the status quo on benefits, costs, and risks.
- C. A project charter authorizing the project manager.
- D. A detailed work breakdown structure and schedule.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A fall-prevention program with bed alarms at a regional medical center costs \$80,000 in its first year. The CFO, who joined from a larger system, estimates it will avoid \$120,000 in first-year injury costs. The center has 14 inpatient units. What is the first-year ROI?

- A. 50%
- B. \$40,000
- C. 150%
- D. About 33%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A project costs \$200,000 and is expected to return an ROI of 25%. What total benefit must it produce?

- A. \$250,000
- B. \$50,000
- C. \$800,000
- D. \$160,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A clinic plans to buy an automated medication cabinet for \$150,000. It is expected to save \$75,000 a year in staff time and waste, but its service contract costs \$15,000 a year. The vendor says the cabinet should last 7 years and is used by 40 other clinics in the state. What is the payback period?

- A. 7 years
- B. 2.5 years
- C. 2.0 years
- D. 0.4 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A \$200,000 investment has a payback period of exactly 4 years. What are its annual net savings?

- A. It cannot be found without a discount rate.
- B. \$800,000
- C. \$200,000
- D. \$50,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A telehealth expansion that the board chair personally favors costs \$100,000 today and will produce net cash inflows of \$60,000 at the end of year 1 and \$60,000 at the end of year 2. The organization uses a 10% discount rate and has 2,000 employees. What is the NPV, and should it proceed on financial grounds?

- A. About \$104,132; proceed.
- B. \$20,000; proceed.
- C. About \$4,132; proceed, because NPV is positive.
- D. About -\$4,132; reject.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A nurse manager is mapping stakeholders for a unit discharge-education project. The system CFO has final budget authority but little interest in unit-level teaching projects. The CFO is known for running marathons and sits on two nonprofit boards. How should the manager engage the CFO?

- A. Keep satisfied: brief updates focused on cost and risk, with escalation when decisions are needed.
- B. Monitor only, with minimal contact.
- C. Keep informed through the full project newsletter.
- D. Manage closely with weekly detailed meetings.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which stakeholder should be managed closely on a project to redesign nurse handoffs?

- A. Every stakeholder, equally.
- B. The chief nursing officer, who controls resources and is eager to see the new process.
- C. Frontline nurses on another unit who are curious but have no decision role.
- D. The hospital's outside auditor, who has authority but little interest in handoffs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A home health agency with an excellent employee retention record learns that a national chain will open a competing agency in its county next year. The agency's founder is retiring soon and its office was renovated last spring. In a SWOT analysis, where does the national chain's entry belong?

- A. Weakness.
- B. Strength, because of its strong retention.
- C. Threat.
- D. Opportunity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which factor is a weakness in a hospital's SWOT analysis?

- A.** A new federal grant program for rural hospitals.
- B.** A highly rated stroke program.
- C.** A new state reimbursement cut.
- D.** An outdated scheduling system the hospital owns and runs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A proposed bedside barcode scanning system fits the budget, and the vendor's software is proven at 200 hospitals. The project lead drives a hybrid car and has a strong IT background. Nurses on the pilot unit say the scanners add steps during medication passes and they plan to work around them. Which type of feasibility is most in question?

- A.** Technical feasibility.
- B.** Economic feasibility.
- C.** None; a proven, affordable system is always feasible.
- D.** Operational feasibility.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A surgical unit's hand hygiene compliance, measured by trained observers, is 68%. The organization's target, set by a committee that meets quarterly, is 90%. A neighboring hospital reports 85%. What gap should the business case address?

- A.** 158 percentage points.
- B.** 90%
- C.** 22 percentage points.
- D.** 17 percentage points.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A nurse leader at a 300-bed community hospital, ~~which recently repainted its main lobby,~~ wants to propose a remote patient monitoring program. The executive team asks her to show whether the program is worth starting at all, compared with keeping current practice. ~~The hospital's foundation holds an annual gala in May.~~ Which document should she prepare?

Target word: which document **Decides:** The decision to invest has not been made yet.

A	Camouflage	Answers the gala detail, not the investment decision.
B	Correct	A business case supports the go/no-go decision before a project starts.
C	Adjacent category	A charter is issued after the decision to proceed.
D	Confusion set	These are planning tools used after approval.

2. Answer A Concept 2

A fall-prevention program with bed alarms at a regional medical center costs \$80,000 in its first year. ~~The CFO, who joined from a larger system,~~ estimates it will avoid \$120,000 in first-year injury costs. ~~The center has 14 inpatient units.~~ What is the first-year ROI?

Target word: ROI percent **Decides:** Net benefit \$40,000 divided by cost \$80,000.

A	Correct	$(\$120,000 - \$80,000) / \$80,000 = 50\%$.
B	Confusion set	That is net benefit in dollars, not a return percentage.
C	Dropped element	Divides benefit by cost without subtracting the cost first.
D	Inverted operation	Divides net benefit by the benefit instead of by the cost.

3. Answer A Concept 2 · transfer

A project costs \$200,000 and is expected to return an ROI of 25%. What total benefit must it produce?

Target word: solve for benefit **Decides:** $\text{Benefit} = \text{cost} \times (1 + \text{ROI})$.

A	Correct	$\$200,000 \times 1.25 = \$250,000$.
B	Dropped element	That is the net gain only, not the total benefit.
C	Confusion set	Divides cost by 0.25, treating ROI as cost's share of benefit.
D	Inverted operation	Divides by 1.25 instead of multiplying.

4. Answer B Concept 3

A clinic plans to buy an automated medication cabinet for \$150,000. It is expected to save \$75,000 a year in staff time and waste, but its service contract costs \$15,000 a year. ~~The vendor says the cabinet should last 7 years and is used by 40 other clinics in the state.~~ What is the payback period?

Target word: payback in years **Decides:** Net annual savings = \$75,000 – \$15,000 = \$60,000.

A	Camouflage	The equipment's useful life, not the payback period.
B	Correct	$\$150,000 / \$60,000 = 2.5 \text{ years}$.
C	Dropped element	Uses gross savings and ignores the service contract.
D	Inverted operation	Divides savings by investment.

5. Answer D Concept 3 • transfer

A \$200,000 investment has a payback period of exactly 4 years. What are its annual net savings?

Target word: solve for annual savings **Decides:** Savings = investment / payback years.

A	Absolute language	Simple payback does not use discounting.
B	Inverted operation	Multiplies instead of dividing.
C	Wrong stem number	The investment, not the yearly savings.
D	Correct	$\$200,000 / 4 = \$50,000$.

6. Answer C Concept 4

A telehealth expansion ~~that the board chair personally favors~~ costs \$100,000 today and will produce net cash inflows of \$60,000 at the end of year 1 and \$60,000 at the end of year 2. The organization uses a 10% discount rate ~~and has 2,000 employees~~. What is the NPV, and should it proceed on financial grounds?

Target word: NPV and decision **Decides:** Discount each inflow at 10% before subtracting the investment.

A	Confusion set	That is the present value of inflows, before subtracting the investment.
B	Dropped element	Adds cash flows without discounting them.
C	Correct	$\$60,000 / 1.10 + \$60,000 / 1.21 \approx \$54,545 + \$49,587 = \$104,132; - \$100,000 = \$4,132$.
D	Inverted operation	Subtracts in the wrong order; the present value of inflows exceeds the cost.

7. Answer A Concept 5

A nurse manager is mapping stakeholders for a unit discharge-education project. The system CFO has final budget authority but little interest in unit-level teaching projects. ~~The CFO is known for running marathons and sits on two nonprofit boards~~. How should the manager engage the CFO?

Target word: engagement strategy **Decides:** High power, low interest.

A	Correct	High-power, low-interest stakeholders are kept satisfied without overloading them.
B	Dropped element	Ignores the CFO's high power over the budget.
C	Confusion set	Keep informed is for low-power, high-interest stakeholders.
D	Adjacent category	Close management is for high-power, high-interest stakeholders.

8. Answer B Concept 5 • transfer

Which stakeholder should be managed closely on a project to redesign nurse handoffs?

Target word: reverse: stakeholder from strategy **Decides:** High power and high interest.

A	Absolute language	The grid exists to tailor effort by power and interest.
B	Correct	High power and high interest calls for close management.
C	Adjacent category	Low power, high interest: keep informed.
D	Confusion set	High power, low interest: keep satisfied.

9. Answer C Concept 6

A home health agency ~~with an excellent employee retention record~~ learns that a national chain will open a competing agency in its county next year. The agency's founder is retiring soon and ~~its office was renovated last spring~~. In a SWOT analysis, where does the national chain's entry belong?

Target word: SWOT category **Decides:** An outside competitor is an external factor.

A	Confusion set	Weaknesses are internal; the competitor is outside the agency.
B	Camouflage	Answers the retention detail, which is a separate internal factor.
C	Correct	An external development that could hurt the agency is a threat.
D	Inverted operation	The competitor's entry is harmful, not favorable.

10. Answer D Concept 6 · transfer

Which factor is a weakness in a hospital's SWOT analysis?

Target word: reverse: factor from category **Decides:** Internal and unfavorable.

A	Inverted operation	External and favorable is an opportunity.
B	Confusion set	Internal and favorable is a strength.
C	Adjacent category	External and unfavorable is a threat.
D	Correct	Internal and unfavorable is a weakness.

11. Answer D Concept 7

A proposed bedside barcode scanning system fits the budget, and the vendor's software is proven at 200 hospitals. ~~The project lead drives a hybrid car and has a strong IT background.~~ Nurses on the pilot unit say the scanners add steps during medication passes and they plan to work around them. Which type of feasibility is most in question?

Target word: feasibility type **Decides:** Concern is whether users will adopt it in their workflow.

A	Adjacent category	The technology is proven; the concern is use.
B	Dropped element	The stem already says it fits the budget.
C	Absolute language	Cost and technology do not guarantee adoption.
D	Correct	Operational feasibility asks whether people and workflows will adopt the solution.

12. Answer C Concept 8

A surgical unit's hand hygiene compliance, measured by trained observers, is 68%. The organization's target, ~~set by a committee that meets quarterly~~, is 90%. ~~A neighboring hospital reports 85%.~~ What gap should the business case address?

Target word: size of the gap **Decides:** Target 90% – current 68%.

A	Inverted operation	Adds the two values instead of subtracting.
B	Dropped element	States the target, not the gap.
C	Correct	$90\% - 68\% = 22$ percentage points.
D	Wrong stem number	Uses the neighboring hospital's 85% as the target.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Purpose of a business case	A business case is written early to justify whether a project should start: it states the need, compares the proposed option with the status quo on benefits, costs, and risks, and recommends a decision.	Confusing it with the project charter or detailed project plan written after approval.	A leader asking whether to invest at all, before a plan exists.
2	Return on investment	$ROI = (\text{total benefit} - \text{cost}) / \text{cost}$, expressed as a percentage.	Dividing benefit by cost (giving 100% too much) or dividing by benefit.	A project cost and a benefit or savings figure over the same period.
3	Payback period	Payback period = initial investment / annual net cash savings (after ongoing costs); it ignores the time value of money.	Using gross savings before ongoing costs, or dividing savings by investment.	An up-front cost plus a yearly savings figure.
4	Net present value	NPV = present value of future net cash flows (discounted at the required rate) – initial investment; accept if $NPV > 0$.	Adding cash flows without discounting them.	Cash flows in future years plus a discount rate.
5	Stakeholder power/interest grid	High power, high interest: manage closely; high power, low interest: keep satisfied; low power, high interest: keep informed; low power, low interest: monitor.	Treating every powerful stakeholder as needing close, frequent involvement.	A stakeholder's authority and level of interest in the project.
6	SWOT: internal vs. external	Strengths and weaknesses are internal to the organization; opportunities and threats are external conditions it does not control.	Calling an outside competitor a weakness or an internal skill an opportunity.	Whether the factor sits inside or outside the organization.
7	Types of feasibility	Economic (worth the cost), technical (can the technology do it), operational (will people and workflows adopt it), schedule (can it be done in time), and legal/regulatory feasibility are assessed separately.	Assuming a project is feasible because it is affordable and the technology works.	A concern about staff use, timeline, law, or technology.
8	Gap analysis	The gap is the difference between the desired future state and the current state, measured with the same metric; it defines the need the project must close.	Using the target alone or the current value alone as the gap.	A baseline measure and a target measure.

Course scope (public description): Business Case Analysis for Healthcare Improvement provides learning experiences that help students develop essential skills for proposing changes that improve and enhance healthcare outcomes. In this course, students will develop a business case during the early stages of a project by assessing the need for the project and the feasibility of initiating a project. Understanding the techniques used to

develop a business case will provide students with the skills to obtain buy-in from key stakeholders and determine the best value strategy. Writing a strong business case presents the benefits, challenges, costs, and risks of moving forward with the project or maintaining status quo. It compares the current situation to a future vision so key stakeholders can make data-driven decisions to move forward with the project. During the development of a business case in this course, students will collaborate with internal and external stakeholders to initiate a healthcare improvement project (HIP) that is grounded in project management principles and influenced by stakeholder perspectives. This course is eligible for an In Progress grade. Please see the Grading Scale Policy for more information.